Data low QC

Abinet Joan

2024-08-30 16:17:17 +0200

Contents

Loading packages 1

Loading data 1

QC 2

Normalisation + scalin 2

Dimension reduction 3

Umap 3

Loading packages

```
suppressMessages(library(dplyr))
suppressMessages(library(Seurat))
suppressMessages(library(patchwork))
suppressMessages(library(ggplot2))
suppressMessages(library(writexl))
suppressMessages(library(SingleR))
suppressMessages(library(scuttle))
suppressMessages(library(scRNAseq))
suppressMessages(library(RColorBrewer))
suppressMessages(library(stringr))
suppressMessages(library(formatR))
```

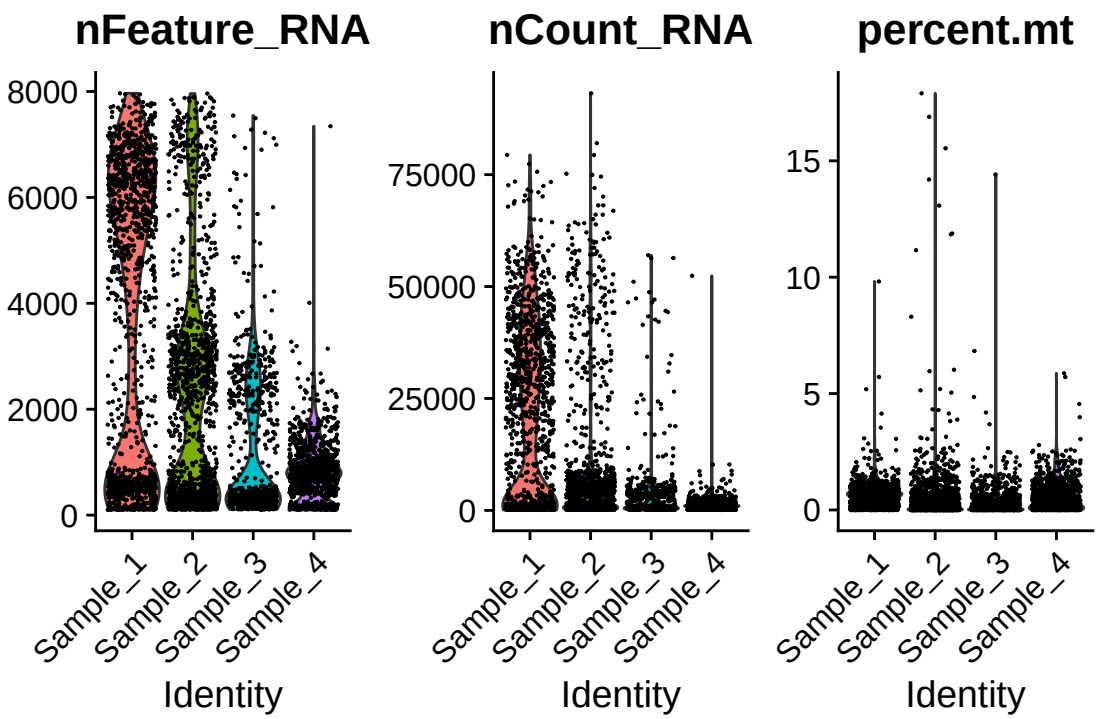
Loading data

```
setwd("../10X-1-Processing")
# The path should be modifie to include your directory of the cellranger output
all_dirs <- dir(path = "Data/outs/per_sample_outs/", full.names = T)

list_sample <- list()
for (i in 1:length(all_dirs)) {

  # here the file path describe the filtered matrix from cellranger output
  Seq_raw_file <- Read10X(data.dir =
paste0(all_dirs[i],"/count/sample_raw_feature_bc_matrix/"))
```

```
  Seurat_file <- CreateSeuratObject(counts = Seq_raw_file, project =
str_sub(all_dirs[i],-8,-1), min.cells = 3, min.features =   100)
  list_sample <- append(list_sample, Seurat_file)
}

lavage_cells10x <- merge(list_sample[[1]], y = list_sample[-1], add.cell.ids =
c("1","2","3", "4"), project = "bronchoalveolar_cells")
```

QC

```
lavage_cells10x[["percent.mt"]] <- PercentageFeatureSet(lavage_cells10x, pattern =
"^MT-")# MT : human cells
lavage_cells10x <- subset(lavage_cells10x, subset = nFeature_RNA < 8000 & percent.mt <
20)# or 5
VlnPlot(lavage_cells10x, features = c("nFeature_RNA", "nCount_RNA", "percent.mt"),
group.by = "orig.ident", ncol = 3, pt.size = 0.1)
```




```
#ggsave("VlnPlot_QC.png", height = 8, width = 12)
```

Normalisation + scalin

```
lavage_cells10x <- NormalizeData(lavage_cells10x, normalization.method = "LogNormalize",
scale.factor = 10000)
lavage_cells10x <- FindVariableFeatures(lavage_cells10x, selection.method = "vst",
nfeatures = 2000)

all.genes <- rownames(lavage_cells10x)
```

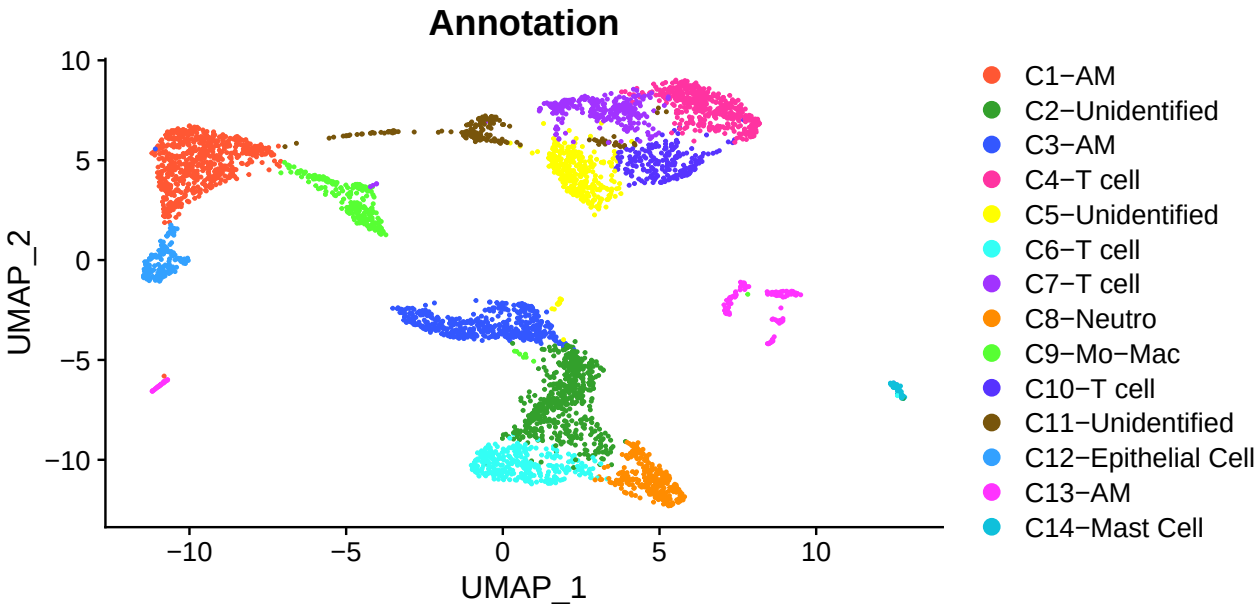
```
lavage_cells10x <- ScaleData(lavage_cells10x, features = all.genes)
```

Dimension reduction

```
lavage_cells10x <- RunPCA(lavage_cells10x, features = VariableFeatures(object =
lavage_cells10x))
#DimPlot(lavage_cells10x, reduction = "pca")

lavage_cells10x <- FindNeighbors(lavage_cells10x, dims = 1:16)
lavage_cells10x <- FindClusters(lavage_cells10x, resolution = 0.5)
lavage_cells10x <- RunUMAP(lavage_cells10x, dims = 1:16)
```

Umap

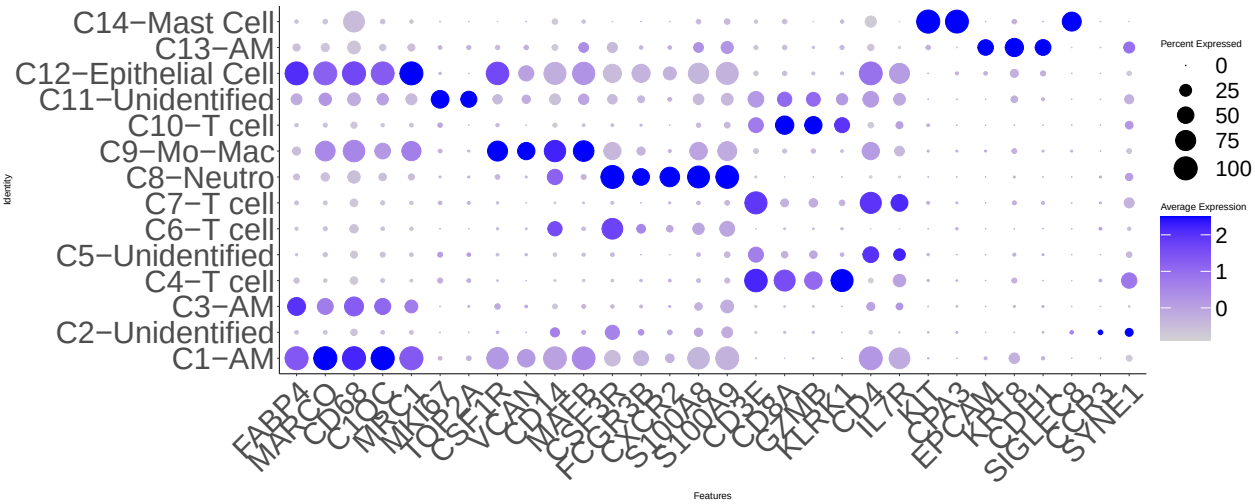
```
lavage_cells10x$Annotation <- lavage_cells10x$seurat_clusters
lavage_cells10x$Annotation <- as.factor(lavage_cells10x$Annotation)
levels(lavage_cells10x$Annotation) <- c("C1-AM", "C2-Unidentified", "C3-AM", "C4-T cell",
"C5-Unidentified", "C6-T cell", "C7-T cell", "C8-Neutro", "C9-Mo-Mac", "C10-T cell",
"C11-Unidentified", "C12-Epithelial Cell", "C13-AM", "C14-Mast Cell")

color_pallette <- c("#FF5733", "#33A02C", "#3357FF", "#FF33A1", "yellow", "#33FFF5",
"#A133FF", "#FF8C00", "#57FF33", "#5733FF", "#78550b", "#33A1FF", "#FF33FF", "#0fc0db")

DimPlot(lavage_cells10x, reduction = "umap", group.by = "Annotation", cols =
color_pallette)
```




```
Marker_gene <- c("FABP4", "MARCO", "CD68", "C1QC", "MRC1", "MKI67", "TOP2A", "CSF1R",
"VCAN", "CD14",
"MAFB", "CSF3R", "FCGR3B", "CXCR2", "S100A8", "S100A9", "CD3E", "CD8A", "GZMB",
"KLGR1", "KLRK1", "CD4", "IL7R", "KIT", "CPA3", "EPCAM", "KRT8", "CDH1", "SIGLEC8",
"CCR3",
```

```
"SYNE1")

DotPlot(lavage_cells10x, features = Marker_gene, group.by = "Annotation") +
theme_classic() + scale_size(range = c(0,12)) + theme(axis.text.x = element_text(angle =
45, hjust = 1), axis.text = element_text(size = 35),
           legend.text   = element_text(size = 25), legend.key.height= unit(1, 'cm'),
        legend.key.width= unit(2, 'cm'))
```

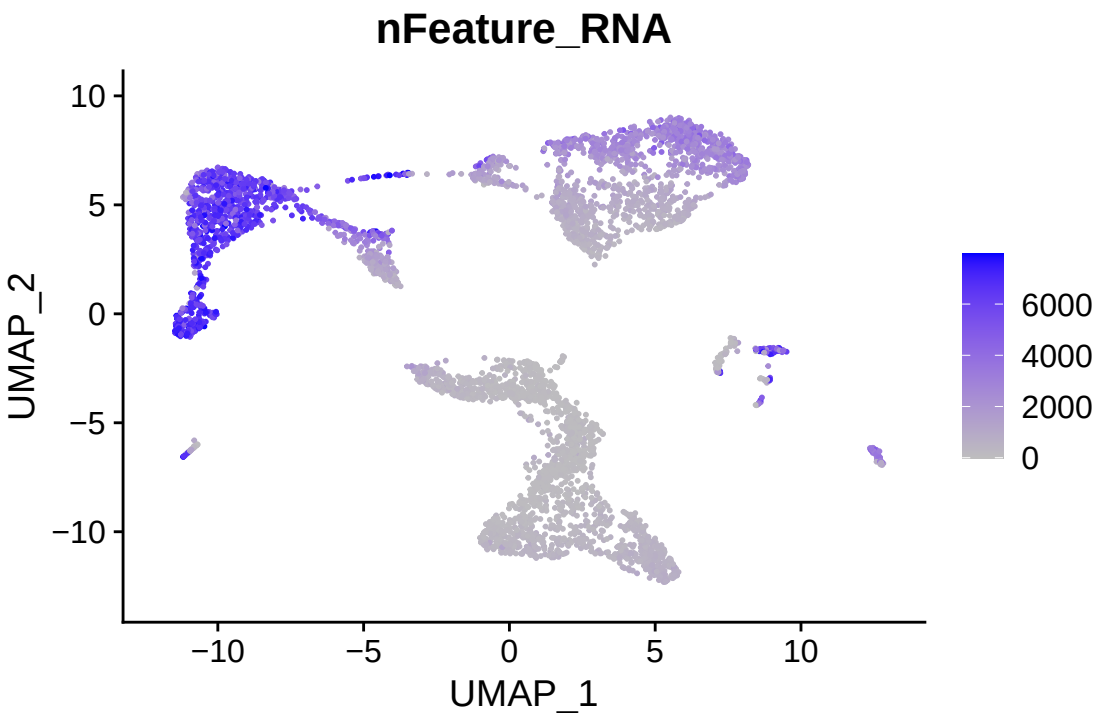
```
## Warning in FetchData.Seurat(object = object, vars = features, cells = cells):
## The following requested variables were not found: KLGR1

## Scale for size is already present.
## Adding another scale for size, which will replace the existing scale.
```




```
#ggsave("plot_lastMinute/Dotplot_10X.pdf", width = 35, height = 10)
```

```
FeaturePlot(lavage_cells10x, features = c("nFeature_RNA"))+ scale_color_gradient(low =
"grey", high = "blue", limit = c(0,7960) )
```

```
## Scale for colour is already present.
## Adding another scale for colour, which will replace the existing scale.
```




```
#ggsave("nFeature.pdf", width = 8, height = 4)
```

```
sessionInfo()
```

```
## R version 4.3.3 (2024-02-29)
## Platform: x86_64-pc-linux-gnu (64-bit)
## Running under: Ubuntu 22.04.4 LTS
##
## Matrix products: default
## BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
## LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.20.so;  LAPACK version 3.10.0
##
## locale:
##  [1] LC_CTYPE=en_US.UTF-8       LC_NUMERIC=C
##  [3] LC_TIME=fr_BE.UTF-8        LC_COLLATE=en_US.UTF-8
##  [5] LC_MONETARY=fr_BE.UTF-8    LC_MESSAGES=en_US.UTF-8
##  [7] LC_PAPER=fr_BE.UTF-8       LC_NAME=C
##  [9] LC_ADDRESS=C               LC_TELEPHONE=C
## [11] LC_MEASUREMENT=fr_BE.UTF-8 LC_IDENTIFICATION=C
##
## time zone: Europe/Brussels
## tzcode source: system (glibc)
##
## attached base packages:
## [1] stats4    stats     graphics  grDevices utils     datasets  methods
## [8] base
##
## other attached packages:
##  [1] formatR_1.14                stringr_1.5.0
##  [3] RColorBrewer_1.1-3          scRNAseq_2.14.0
##  [5] scuttle_1.10.1              SingleCellExperiment_1.22.0
```

```
##  [7] SingleR_2.2.0               SummarizedExperiment_1.30.2
##  [9] Biobase_2.60.0              GenomicRanges_1.52.0
## [11] GenomeInfoDb_1.36.0         IRanges_2.34.0
## [13] S4Vectors_0.38.1            BiocGenerics_0.46.0
## [15] MatrixGenerics_1.12.2       matrixStats_1.0.0
## [17] writexl_1.4.2               ggplot2_3.4.2
## [19] patchwork_1.1.2             SeuratObject_4.1.3
## [21] Seurat_4.3.0                dplyr_1.1.2
##
## loaded via a namespace (and not attached):
##   [1] ProtGenerics_1.32.0           spatstat.sparse_3.0-2
##   [3] bitops_1.0-7                  httr_1.4.6
##   [5] tools_4.3.3                   sctransform_0.3.5
##   [7] utf8_1.2.3                    R6_2.5.1
##   [9] lazyeval_0.2.2                uwot_0.1.16
##  [11] withr_2.5.0                   sp_2.0-0
##  [13] prettyunits_1.1.1             gridExtra_2.3
##  [15] progressr_0.13.0              cli_3.6.1
##  [17] spatstat.explore_3.2-1        labeling_0.4.2
##  [19] spatstat.data_3.0-1           ggridges_0.5.4
##  [21] pbapply_1.7-2                 Rsamtools_2.16.0
##  [23] R.utils_2.12.2                parallelly_1.36.0
##  [25] rstudioapi_0.14               RSQLite_2.3.1
##  [27] generics_0.1.3                BiocIO_1.10.0
##  [29] ica_1.0-3                     spatstat.random_3.1-5
##  [31] Matrix_1.6-1                  ggbeeswarm_0.7.2
##  [33] fansi_1.0.4                   abind_1.4-5
##  [35] R.methodsS3_1.8.2             lifecycle_1.0.3
##  [37] yaml_2.3.7                    BiocFileCache_2.8.0
##  [39] Rtsne_0.16                    grid_4.3.3
##  [41] blob_1.2.4                    promises_1.2.0.1
##  [43] ExperimentHub_2.8.0           crayon_1.5.2
##  [45] miniUI_0.1.1.1                lattice_0.22-5
##  [47] beachmat_2.16.0               cowplot_1.1.1
##  [49] GenomicFeatures_1.52.0        KEGGREST_1.40.0
##  [51] pillar_1.9.0                  knitr_1.43
##  [53] rjson_0.2.21                  future.apply_1.11.0
##  [55] codetools_0.2-19              leiden_0.4.3
##  [57] glue_1.6.2                    data.table_1.14.8
##  [59] vctrs_0.6.3                   png_0.1-8
##  [61] spam_2.9-1                    gtable_0.3.3
##  [63] cachem_1.0.8                  xfun_0.39
##  [65] S4Arrays_1.2.1                mime_0.12
##  [67] survival_3.5-8                interactiveDisplayBase_1.38.0
##  [69] ellipsis_0.3.2                fitdistrplus_1.1-11
##  [71] ROCR_1.0-11                   nlme_3.1-164
##  [73] bit64_4.0.5                   progress_1.2.2
##  [75] filelock_1.0.2                RcppAnnoy_0.0.21
##  [77] irlba_2.3.5.1                 vipor_0.4.5
##  [79] KernSmooth_2.23-22            colorspace_2.1-0
##  [81] DBI_1.1.3                     ggrastr_1.0.2
##  [83] tidyselect_1.2.0              bit_4.0.5
##  [85] compiler_4.3.3                curl_5.0.1
##  [87] xml2_1.3.4                    DelayedArray_0.26.3
```

```
##  [89] plotly_4.10.2                rtracklayer_1.60.0
##  [91] scales_1.2.1                 lmtest_0.9-40
##  [93] rappdirs_0.3.3               digest_0.6.33
##  [95] goftest_1.2-3                spatstat.utils_3.0-3
##  [97] rmarkdown_2.23               XVector_0.40.0
##  [99] htmltools_0.5.5              pkgconfig_2.0.3
## [101] sparseMatrixStats_1.12.0     highr_0.10
## [103] dbplyr_2.3.2                 fastmap_1.1.1
## [105] ensembldb_2.24.0             rlang_1.1.1
## [107] htmlwidgets_1.6.2            shiny_1.7.4.1
## [109] DelayedMatrixStats_1.22.1    farver_2.1.1
## [111] zoo_1.8-12                   jsonlite_1.8.7
## [113] BiocParallel_1.34.2          R.oo_1.25.0
## [115] BiocSingular_1.16.0          RCurl_1.98-1.12
## [117] magrittr_2.0.3               GenomeInfoDbData_1.2.10
## [119] dotCall64_1.0-2              munsell_0.5.0
## [121] Rcpp_1.0.11                  reticulate_1.30
## [123] stringi_1.7.12               zlibbioc_1.46.0
## [125] MASS_7.3-60.0.1              AnnotationHub_3.8.0
## [127] plyr_1.8.8                   parallel_4.3.3
## [129] listenv_0.9.0                ggrepel_0.9.3
## [131] deldir_1.0-9                 Biostrings_2.68.1
## [133] splines_4.3.3                tensor_1.5
## [135] hms_1.1.3                    igraph_1.5.0.1
## [137] spatstat.geom_3.2-4          reshape2_1.4.4
## [139] biomaRt_2.56.1               ScaledMatrix_1.8.1
## [141] BiocVersion_3.17.1           XML_3.99-0.14
## [143] evaluate_0.21                BiocManager_1.30.21
## [145] httpuv_1.6.11                RANN_2.6.1
## [147] tidyr_1.3.0                  purrr_1.0.1
## [149] polyclip_1.10-4              future_1.33.0
## [151] scattermore_1.2              rsvd_1.0.5
## [153] xtable_1.8-4                 restfulr_0.0.15
## [155] AnnotationFilter_1.24.0      later_1.3.1
## [157] viridisLite_0.4.2            tibble_3.2.1
## [159] memoise_2.0.1                beeswarm_0.4.0
## [161] AnnotationDbi_1.62.1         GenomicAlignments_1.36.0
## [163] cluster_2.1.6                globals_0.16.2
```